

1. Preprocessing & Windowing

DASIG Corpus: 60 Subjects $\times$ 3 Trials
65 Sensor Channels @ 200 Hz



Sliding Window: 0.5 s (100 steps, 50% overlap)
Per-Trial Z-Score: $x_{\text{norm}} = (x - \mu) / \sigma$



Window-Level Labeling: $y = \max(t_{\text{labels}})$
69,094 Windows (87.6% SAFE / 12.4% DANGER)



2. Dynamic Kinematic Feature Expansion

Position (Raw): $X \in \mathbb{R}^{(65 \times 100)}$

Velocity (1st Deriv): $V = dX/dt \in \mathbb{R}^{(65 \times 100)}$

Acceleration (2nd Deriv): $A = d^2X/dt^2 \in \mathbb{R}^{(65 \times 100)}$



Expanded Kinematic Tensor: $[X; V; A]$
Dimensions: (B, 195, 100)